the task ✗ ○

a compound
and its molar dose

the strain: a deletion,
or the bAID host with none

cell graph transformer
one compound token in front of
the gene tokens; the genes attend to it

read at the deleted genes:
the deletion profile
(the public screens)

read at the mean over genes:
host growth and fitness
(the bAID wet-lab readout)

a mixture is a SET of
compound tokens, each
modulated by its own
log-molar dose
(proposed, not run)

the data ✗ ●

the pooled chemogenomic table: 4 screens, 6,042,771 cells,
5,463 compounds over 5,863 genes; 5,585,863 cells carry a molar dose
Vanacloig 2022, Hillenmeyer HET, Hoepfner 2014, Wildenhain 2015

the private Bioscreen runs of 2021: 977 wells, 6 inhibitors, one strain
ex21 titrations (180 wells), ex23 combinations (197 wells, 63 of them),
3 isobole grids (600 wells)

the evaluation ✗ ○

public screens: compound-cold folds,
the per-compound centered Spearman
against each compound's reliability ceiling

→

wet lab: growth over 63 combinations (AUROC)
and fitness over the 21 that grew (Spearman),
against Bliss (0.87) and Loewe (0.96)

→

the industrial reading:
rank the worst combination
scenarios of a hydrolysate